PerplexityAI ~ Learning and applying Active Inference & Reading the 2022 Textbook ~ July 31, 2023

> **Source**: [YouTube] **Playlist**: Active Inference Journal **Duration**: 13:55 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

it's July 31st 2023 this is Daniel and this video is going to be using perplexity.ai which is an AI based conversational search engine using perplexity.ai to learn and apply active inference pretty fun conversation that I just had a lot of epistemic and pragmatic value so I started by using the new feature of beta from perplexity AI today which was the ability to upload a file with several other language models increasingly have and I uploaded the 2022 active inference textbooks textbook and I first aim for defenses write an elaborate PhD dissertation that summarizes all domain aspects and technical details of the active inference textbook attached here's the PDF and it didn't want to immediately go for it but it gave some summary but I was looking for something a lot deeper than just an overall summary of the book so I asked what is the free energy principle and how is it related to active inference it's an interesting answer but I'm going to keep on moving because there's some really detailed things it gets to what is expected free energy give multiple explanations including the perspective of math science and programming now here I utilized The Limited number of copilot runs five every four hours so this engaged a sequential process of searching the web finding sources different papers and YouTube channels videos and writing about broadly the aspects of expected free energy epistemic value and here extrinsic value but usually called pragmatic value so it's a pretty good representation so I asked for more of an application explain what epistemic value and pragmatic value would represent in the case of someone who wanted to turn on their air conditioner to prevent overheating however first had to find a key that would unlock the air conditioner's control panel again used copilot search the web looked at a bunch of sources clarified and improved its understanding of epistemic and pragmatic value and relatively accurately describing the epistemic value this represents the agent's uncertainty about the environment and the need to gather information to reduce that uncertainty in this scenario the epistemic values associated with the agent's uncertainty about the location of the key by searching for the key the agent reduces their uncertainty and gains knowledge about the environment which is considered valuable from an epistemic perspective pragmatic value this represents the agent's preferences or goals which in this case is to turn on the air conditioner to prevent overheating in the context of active inference the agent's goal is to minimize the expected free energy by finding the optimal balance between epistemic and pragmatic values in this scenario the agent needs to gather information about the key's location epistemic value and take actions to turn on the air conditioner pragmatic value to achieve their goal and prevent overheating again with copilot I asked right comprehensive professional python style pseudocode to fully simulate the situation with the person air conditioner and key it asked for more information on what size I wanted the world to be I said 20 by 20. it looked up some related terms discovered some agent-based modeling repos wrote a python style script I didn't run it I just kept going great can you update the script to explicitly utilize open source active inference implementations for example using Pi mdp Pi mdp could be used to specify the generative model for the human agent develop that as much as you can and then I will be ready to provide more technical information after it asks what is what should the generative model container do I said the person should have three hidden physiological states too cold just right too hot the ability to observe temperature values on a 100 scale and the ability to move in the grid within a three by three adjacency Matrix there was some searching there was the discovery of multiple papers kaggle archive NIH research Gates two of the model stream Pi mdp videos from reactive inference Institute and it specified the problem giving what is a broadly compatible but not complete or coherence Pi mdp implementation for for a number of reasons but again it's a pseudo code and it gives a disclaimer it's a high level representation of the implementation and you may need to adapt your specific requirements and the pi mdp's Library Syntax for more information on using pi mdp you can refer to the official documentation and the step-by-step tutorial on active inference so those pieces of information as PDFs or as websites probably eventually or currently could be already brought in to the model I said keep working with that remember that action selection that the person takes needs to consider the variational free energy of sensory States the temperature and their hidden internal States physiological temperature copilot did a search implemented the variational free energy now what was very interesting was in the code it actually computed the expected free energy and I don't think it did variational at all in this updated pseudocode we added the computation of the expected free energy using the agent compute expected free energy method the action selection is now based on the efe which considers both sensory States temperature and hidden internal States physiological temperature the agent moves in the grid within a three by three adjacency Matrix and unlocks the air conditioner when it reaches the air conditioner and has the key I said we're not quite unexpected free energy yet can you rewrite the script with proper calculation of variational free energy approximating the model with the lowest price which is the highest model evidence apologies for the confusion there was some infusion apparently I can provide you with a modified pseudocode that incorporates the concept of variational free energy based on the principles of active inference great more code more provocative code a variational free energy threshold-based calculations well compute vfv is introduced this is just pseudocode you can Implement your specific vfv calculation within this function okay that sounds good please note you'll need to replace the placeholder code and compute the IP function with your actual vfp calculation based on your specific model and requirements so again keeping in mind that the textbook is in the window or something write all the specifications and considerations that would be important in order to write the compute vfv function based upon textbook equation 2.5 and anywhere else in the book that is relevant but variational free energy again probably not 100 percent accurate for example ICD is set here to be the observation Matrix but that should be the prior but in general it's it's more accurate than not so I went a little broader with give 11 different perspectives or interpretations about what variational free energy is think broadly we're Global and intergenerational textbook group 11 different perspectives could argue whether they're all within a kind of technical or scientific domain but but let's see if it can go broader that's awesome can you comprehensively describe expected free energy equation 2.6 in the textbook expected free energy time Horizon action selection uncertainty adaptation computational complexity model selection overall expected free energy extends the concept of variational free energy to incorporate the expected future surprise or prediction error it provides a useful measure for guiding the agent's action selection and adaptation in Dynamic and uncertain environments 100 true give 15 Salient concise provocative points that convey expected free energy to someone interested in embodied embedded inactive approaches 15 points if you were to derive a word list of the 17 most important terms to know from the textbook what would it be I'm thinking about words that are rare outside of this textbook and overrepresented in their syntactic abundance and semantic Central centrality in this textbook here are the 17 Salient and rare terms from a textbook that are overrepresented in their syntactic abundance and semantic centrality here they are active inference variational free energy expected free energy generative model Bayesian inference prediction error surprise self-evidencing normative framework message passing neurobiology computational models procedural knowledge belief updating sentient behavior and activism unified theory these terms are Central to the concepts and theories presented in the textbook and are important for understanding the principles and implications of active inference they provide a unique perspective on how we perceive plan and act in the world and how we can use computational models to simulate and model sentient Behavior can you make a table where the rows are those 17 items terms and there's a second column for a synthetic definition based on your best perspective give us a table active inference a normative framework that characterizes Bayes optimal behavior and cognition in living organisms by minimizing the surprise of their sensory observations foreign this is really interesting after this conversation with your help I will be doing some applied active inference modeling something involving what is discussed in chapter 3 chapter 4 chapter 6 and chapter 9. can you lay out a technical agenda for me to completely specify an active inference agent over the next one month I'm ready to work two hours per day with focus and attention on this task and now you are there to catalyze so there's 15 points to consider before doing that kind of work but I didn't like the list so I said can you lay it out like a table with a few clear Milestones that would demonstrate clear progress like for a PhD and I said sure here's the table with a few clear Milestones that would demonstrate clear progress in specifying an active inference agent over the next month Milestone one understanding active inference Milestone 2 study chapter 3 Implement generative models explore chapter four design active inference models chapter 6. active inference industry time active inference and continuous time interestingly it mentions chapter 7 discrete time even though I didn't mention chapter 7. and it linked up six seven and eight and nine and fought it linked up it linked up from four to nine with the Milestone chapter alignment but not aligned for these three to mix it up Milestone 10 Implement and refine models Milestone 11 experiment and evaluate and asked what would someone in your position ask me in partying before I head off to tackle these 11 milestones and more 11 questions to keep in mind so that's all for now thought it was a pretty interesting and informative conversation I applaud technically what perplexity.ai is doing and I look forward to seeing what uh people in the textbook group people who are looking at how to understand the equations and the contexts for different things in the textbook it would be awesome to see how these tools can help us learn and apply all right till next time